Bone Fracture Classification

Hybrid CNN + Vision Transformer

KBG Hackathon 2025 • IIT Mandi

Kamand Bioengineering Group

Problem Statement

Develop an AI system for binary bone fracture classification from X-ray radiographs to assist radiologists in faster, more accurate fracture identification.

Team: Aditya Rai | Domain: Medical Image Analysis

Literature Review & Approach Justification

Literature Review

- CNNs (ResNet, EfficientNet) dominate fracture detection
- → strong local feature extraction from texture/edges
- ViTs capture long-range spatial dependencies via attention
- → whole-bone context for subtle / hairline fractures
- Hybrid CNN+ViT outperforms single-backbone approaches
- → e.g., TransMed, MedViT show complementary strengths
- Cross-attention fusion > simple concatenation
- → allows bidirectional feature interaction

Our Approach

- EfficientNet-B3 (CNN): efficient local features (d=1536)
- ViT-Small-Patch16-224: global self-attention (d=384)
- Cross-Attention Fusion into shared d=512 space
- 2-Phase Transfer Learning:
- Phase 1 → frozen backbones (learn fusion head)
- Phase 2 → full fine-tuning with low LR
- Only ImageNet pre-trained weights (no medical pretrain)
- Designed for clinical decision support

Why Hybrid CNN + ViT?

CNN	Texture, edges, bone cortex patterns → local features
ViT	Whole-bone geometry, fracture line continuity → global context
Fusion	Cross-attention lets CNN & ViT features attend to each other → richer representation
Result	**98.22% test accuracy | 0.9822 Macro F1 | 0.9970 AUC-ROC**

Data Preprocessing & Strategy

Dataset Overview

- Bone Fracture Binary Classification (Kaggle)
- Total images: 10,581
- → Train: 9,246 | Val: 829 | Test: 506
- Classes: Fractured / Not Fractured
- Format: PNG/JPEG, variable resolution
- Pre-split into train/val/test folders

Preprocessing Pipeline

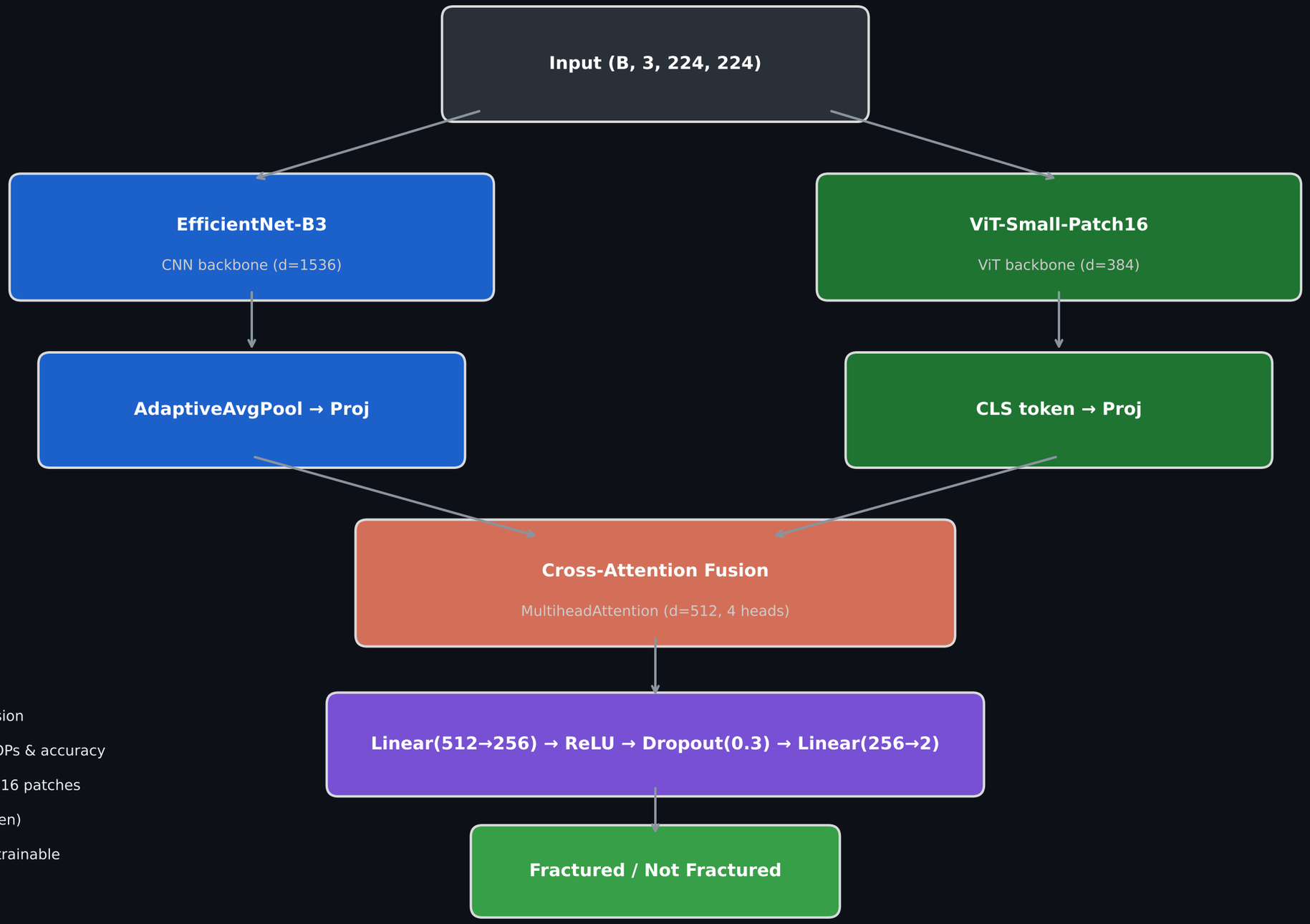
- Resize to 224×224 pixels
- ImageNet normalization (μ, σ per channel)
- Pixel values → [0, 1] → normalized
- Consistent 3-channel RGB conversion
- Reproducible with seed=42
- num_workers=2, pin_memory on CUDA

Data Augmentation Strategy (Training Only)

Augmentation	Parameters	Rationale
Horizontal Flip	p=0.5	Simulate L/R orientation differences
Affine Rotation	±15°	Account for positioning variations
Brightness/Contrast	±20%	Handle exposure differences across X-ray machines
CLAHE	clip=2.0	Enhance local contrast in bone structures
Normalization	ImageNet μ,σ	Standardize for pretrained backbone input

Validation & Test: only resize + normalize (no augmentation)

Model Architecture — HybridCNNViT



Key Design Choices:

- ✦ Cross-attention > concat fusion
- ✦ EfficientNet-B3 balances FLOPs & accuracy
- ✦ ViT-Small: 22M params, 16×16 patches
- ✦ Total: 33.9M params (unfrozen)
- ✦ Phase 1 frozen: only 1.58M trainable

Training Details

Phase 1 — Frozen Backbones

- Epochs: 10 (of total 50 budget)
- Trainable params: 1.58M (fusion + classifier)
- Learning rate: 1e-4 (AdamW)
- Scheduler: CosineAnnealingLR
- Purpose: learn good fusion weights first
- Converged to 97.95% val accuracy

Phase 2 — Full Fine-tuning

- Epochs: 11–15 (early stopped at epoch 15)
- Trainable params: 33.9M (all layers)
- Learning rate: 1e-5 (10× lower)
- Scheduler: CosineAnnealingLR (reset)
- Early stopping: patience=10 on val F1
- Best: epoch 13 → 99.16% val acc, 0.9913 F1

Hyperparameters & Configuration

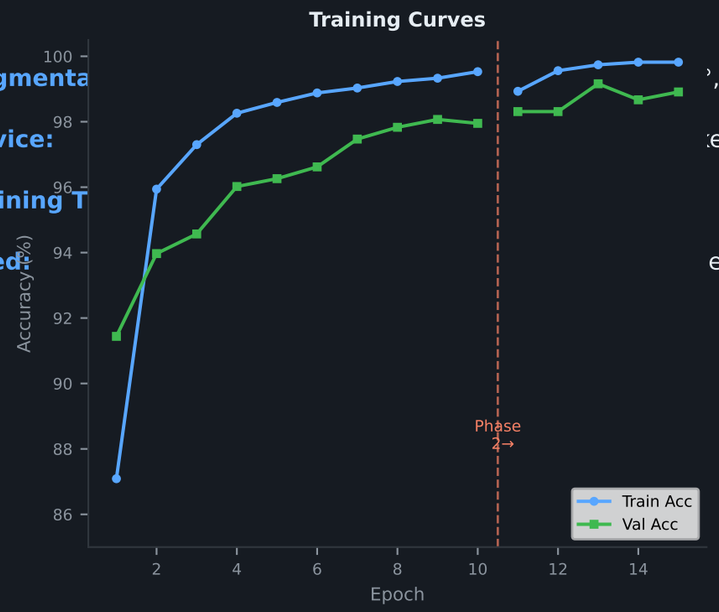
Loss Function:	CrossEntropyLoss
Optimizer:	AdamW (weight_decay=1e-2)
Batch Size:	16
Image Size:	224 × 224

Augmenta ², Brightness, CLAHE

Device: xtend)

Training T

Seed: e)



Results & Performance Metrics

98.22%
Test Accuracy

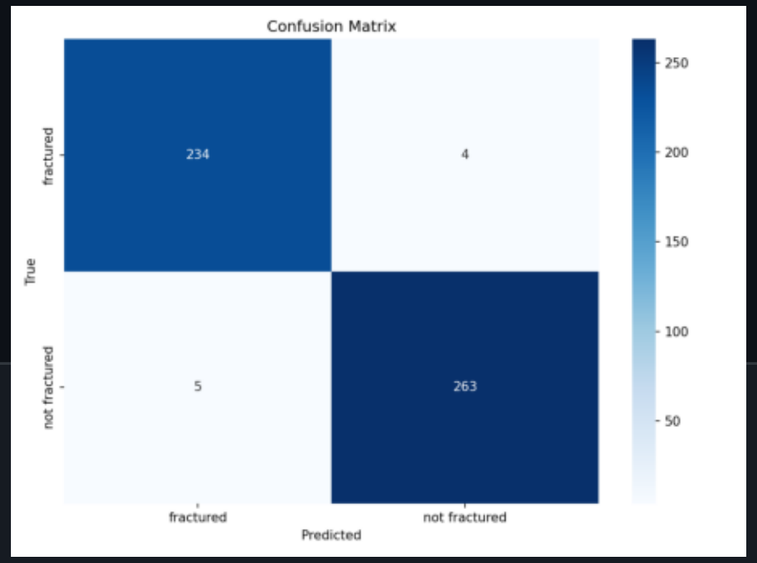
0.9822
Macro F1-Score

0.9970
AUC-ROC

Per-Class Metrics

Class	Precision	Recall	F1-Score	Support
Fractured	0.9791	0.9832	0.9811	238
Not Fractured	0.9850	0.9813	0.9832	268
Macro Avg	0.9820	0.9823	0.9822	506

Confusion Matrix



Additional Metrics

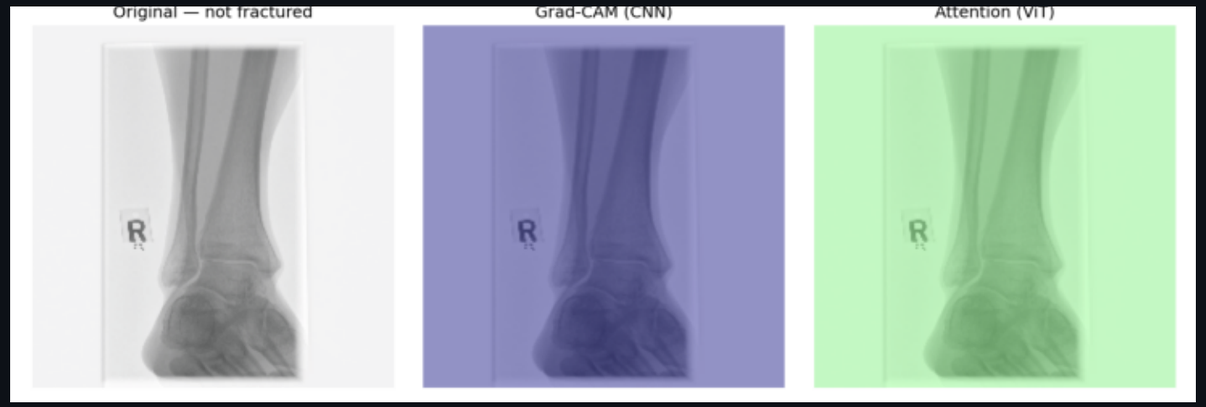
Training Time	Inference Speed	Model Size	
~45 min (Apple M2 MPS)	5.67 ms/image	130 MB	0.9913 (epoch 13)

Explainability — Grad-CAM & Attention Maps

CNN Grad-CAM highlights fracture regions | ViT Attention shows global focus patterns

Fractured — Full Analysis

Not Fractured — Full Analysis



Challenges, Limitations & Future Work

Challenges Faced

- Large model (33.9M params) on limited hardware
- MPS backend lacks full CUDA feature parity
- Some corrupted JPEG images in dataset
- Balancing CNN and ViT learning rates
- Cross-attention fusion hyperparameter tuning
- 2-phase training required careful LR scheduling

Current Limitations

- Binary classification only (fracture / no fracture)
- No fracture type sub-classification
- Single anatomical region generalization unknown
- Not validated on clinical-grade DICOM images
- Limited external dataset validation
- No uncertainty quantification

Future Improvements

→ Multi-class Extension:	Expand to 7+ fracture types (simple, comminuted, spiral, stress, etc.)
→ Uncertainty Estimation:	Monte Carlo Dropout or ensemble for confidence scores
→ Multi-Region Support:	Train on diverse anatomical regions (hand, wrist, hip, spine)
→ DICOM Integration:	Direct DICOM reading with windowing for clinical deployment
→ Model Compression:	Knowledge distillation → lighter model for edge devices
→ Vision Mamba:	Explore state-space models (Vim) for linear-complexity alternatives
→ Clinical Validation:	IRB-approved prospective study with radiologist comparison

Thank you! Questions?